China Semiconductors

China Semis: Huawei's LogicFolding, an underestimated breakthrough

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Huawei unveiled the **"Tau (τ) Scaling Law"** at ISCAS 2026 on May 25, 2026, we wrote about the implications in our previous report. One of the key tech enabler in the Tau Law is called **LogicFolding**, which is used to improve chip performance through stacking. Recent market feedback remains skeptical, but the skepticism appears to underappreciate how different LogicFolding is from existing 3DIC packaging solutions that is already commercialized. **We believe it could be one of the most important breakthroughs in China semis that is underestimated.** This report explains how is the approach is technically differentiated, and why it is impressive.

Skepticism one: LogicFolding is just copying other 3DIC solutions and won't be able to solve the overheat issue after stacking.

The most common criticism: LogicFolding is the same as other leading 3DIC solutions like TSMC SoIC, Intel Foveros Direct, or Samsung 3D Cube. However, if that is the case, then just like these solutions, stacking can easily increase transistor density, but putting two dies on top of each other will inevitably double the power consumption and heat generation, and therefore fail to deliver practical performance gains (actually it will usually lead to lower clock frequency due to overheat). However, spec from Huawei's Kirin 2026 chip showed opposite results: 41% power efficiency gain and 13% frequency improvement. This indicates that LogicFolding is actually a completely different packaging technology. The advance is not merely more transistors per footprint; it is a simultaneous improvement in density, energy efficiency, and frequency that justify itself as the next generation process.

So what is the real differentiation? The key technology breakthrough is, LogicFolding breaks down one functional logic circuits that was originally designed in one die, and distributes that across two vertically stacked wafer layers at the design stage, which is completely different from packaging two independently designed good dies together (usually logic + SRAM).

Why is that important? Because only through this new design, the packaging can reduce the power consumption and increase the frequency. To understand how it works, let's take a step back and think about how logic circuits functions. If we keep it oversimplified, then you can think of the operation of a chip that is basically two parts, the transistors and the interconnects between the transistors. Therefore, power consumption and latency are just the aggregated sum of these two parts. Without node migration, obviously Huawei cannot make any improvement on the transistor level, however they can use the packaging to reduce the power consumption and latency for interconnects. For the traditional 2D design, if two transistors are far away from each other, then talking to each other need to go through a long distance that lead to long latency and high power consumption. But if you use packaging to stack the two transistors on top of each other, the interconnect distance will be much shorter and therefore significantly reducing the power consumption and latency. The Kirin 2026 showed -30% change of wire length in a representative core, which is not yet fully optimized according to their disclosure and thus further improvement could be achieved in next generations.

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Why is this breakthrough impressive? To realize the performance gain demonstrated above, there are three challenges: new EDA tool, new circuit design/optimization methodology, and advanced packaging. This is also why LogicFolding should not be treated as interchangeable with conventional 3DIC solutions. These other solutions can solve the packaging challenge, but they don't have EDA and Fabless innovations that are equally critical to achieve these results. The barrier is therefore not only packaging capability, but also design methodology, software tooling, and architecture-level co-development.

This is exactly the reason why I call the Tau's law another DeepSeek moment. Just like DeepSeek, they are forced to innovate on infrastructure as they are constrained on compute power, which helped to reduce the inferencing cost by an order of magnitude. In this case, Huawei is constrained on lithography, but that indeed forced them to innovate on packaging, which is also very critical for the development of China Semis.

Skepticism two: this solution stays on paper, we can't trust it until we see it.

A second criticism is that LogicFolding is still a research concept that may need years before reaching product commercialization. But the reality is Huawei plans the first commercial LogicFolding implementation in a Kirin smartphone processor scheduled for fall 2026, which is very likely the next Mate series. So it's not years away, it's just months away. If Huawei doesn't have the chip already in volume production, it is hard for us to imaging why they would say this is coming out in next mobile phone. So this tech breakthrough is not just a concept, but already at delivery stage.

Additionally, some may also concern that if the packaging yield is low, then the cost of chip will be so high that it can't really be commercially successful. But we are impressed to see that Huawei claimed that they can achieve ~100% yield through smart redundancy, implying Huawei is already addressing one of the key cost and manufacturability objections associated with this technology. In practice, that means the market can easily see supporting facts in the next few months after product launch, as chip performance benchmark data and teardown evidence should be able to validate whether the performance claims and design novelty are real. Market should re-rate when that happens.

Skepticism three: Global players can easily copy that

A third criticism is that even if LogicFolding works, global leaders can replicate it quickly, limiting Huawei's long-term advantage. That argument overlooks the complexity of this end-to-end optimization of EDA/design/manufacturing. Most global fabless likely won't be willing to adopt the technology until it's matured, but EDA vendors cannot make improvement if customer is not willing to use it, and foundry also cannot further optimize their packaging process around it. Therefore, although technically global players can copy this concept, the implementation will be very challenging. Note that Huawei is not necessarily the first company to create this idea of vertical logic partitioning, but it is the first to reach commercialization of this technology and achieve impressive power consumption reduction and frequency improvement.

On the other hand, even if global vendors can copy Huawei on that technology and pursue similar architectures, early mover advantages in EDA adaptation, design libraries, process integration, and product learning cycles can still matter materially. That would not erase the node gap with frontier foundries, but it could help Huawei offset part of that gap and potentially narrow competitive distance in selected end markets over time.

We like **SMIC, NAURA, Piotech** the most under this theme, as this opens up more upside for advance logic demand in China. SMIC could sell more advanced logic chips, NAURA could benefit from acceleration in advanced logic capacity expansion, and Piotech could potentially benefit from the surging bonding tools demand.

BERNSTEIN TICKER TABLE

			3 Jun 2026		TTM		Reported EPS			Reported P/E (x)		
			Closing	Price	Rel.							
Ticker	Rating	Cur	Price	Target	Perf.	Cur	2025A	2026E	2027E	2025A	2026E	2027E
688981.CH (SMIC-A)	O	CNY	134.00	120.00	10.3%	USD	0.09	0.10	0.13	231.0	197.8	157.1
981.HK (SMIC-H)	O	HKD	82.95	70.00	53.8%	USD	0.09	0.10	0.13	123.4	105.6	83.9
688347.CH (Hua Hong-A)	O	CNY	233.24	140.00	332.4%	USD	0.03	0.11	0.16	N/M	324.1	222.0
1347.HK (Hua Hong-H)	O	HKD	151.70	100.00	343.3%	USD	0.03	0.11	0.16	609.8	181.9	124.6
002371.CH (NAURA)	O	CNY	614.84	680.00	44.4%	CNY	5.66	10.22	16.41	108.6	60.2	37.5
688012.CH (AMEC)	O	CNY	285.92	500.00	86.6%	CNY	3.40	4.95	7.18	84.1	57.8	39.8
688072.CH (Piotech)	O	CNY	615.49	580.00	282.3%	CNY	3.32	8.12	12.40	185.4	75.8	49.6
688256.CH (Cambricon)	O	CNY	1,378.10	2,000.00	187.2%	CNY	(1.09)	5.38	11.26	N/M	256.0	122.4
688041.CH (Hygon)	O	CNY	285.94	280.00	59.3%	CNY	0.83	1.35	2.42	344.5	211.9	118.2
ASIAX			2,056.69									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

688256.CH, 688041.CH base year is 2024;

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

We rate SMIC, Hua Hong, NAURA, AMEC, Piotech, Cambricon, Hygon Outperform benefiting from this new roadmap for China Semis.

On beneficiaries, we see the Tau Law roadmap as structurally positive for China's leading foundry, advanced-packaging supply chain, and AI chip ecosystem. On foundry, **SMIC** is a clear strategic partner and likely a critical enabler if Huawei is to deliver the roadmap and eventually reach TSMC A14-equivalent logic density via multi-die integration without EUV, implying sustained demand for SMIC's most advanced DUV based multi-patterning advanced logic nodes. **Hua Hong** likely will also benefit as the expected acquisition of Huali who is potentially also working on advanced logic packaging. In semicap, we see upside skewed to domestic tool vendors who has higher exposure for advanced logic and packaging: **NAURA** should benefit more than **AMEC** as it has much higher exposure in advanced logic, while **Piotech** has been working on bonding tools for advanced packaging and thus market will perceive it as one of the core beneficiaries. For AI fabless, Huawei's framework provides a clearer path to sustain performance scaling under EUV constraints, which should be supportive for domestic accelerator and CPU vendors such as **Cambricon** and **Hygon** as they co-optimize architectures to exploit higher effective density and lower latency interconnect within China's manufacturing envelope. On the other hand they also face more intense competition from Huawei so the upside will be smaller for the AI fabless names compared to foundry and semicap.

DETAILS

HUAWEI'S ROADMAP ON CIRCUITS AND SYSTEMS

In the event, Huawei released its roadmap for chip improvement under the guidance of Tau Scaling Law.

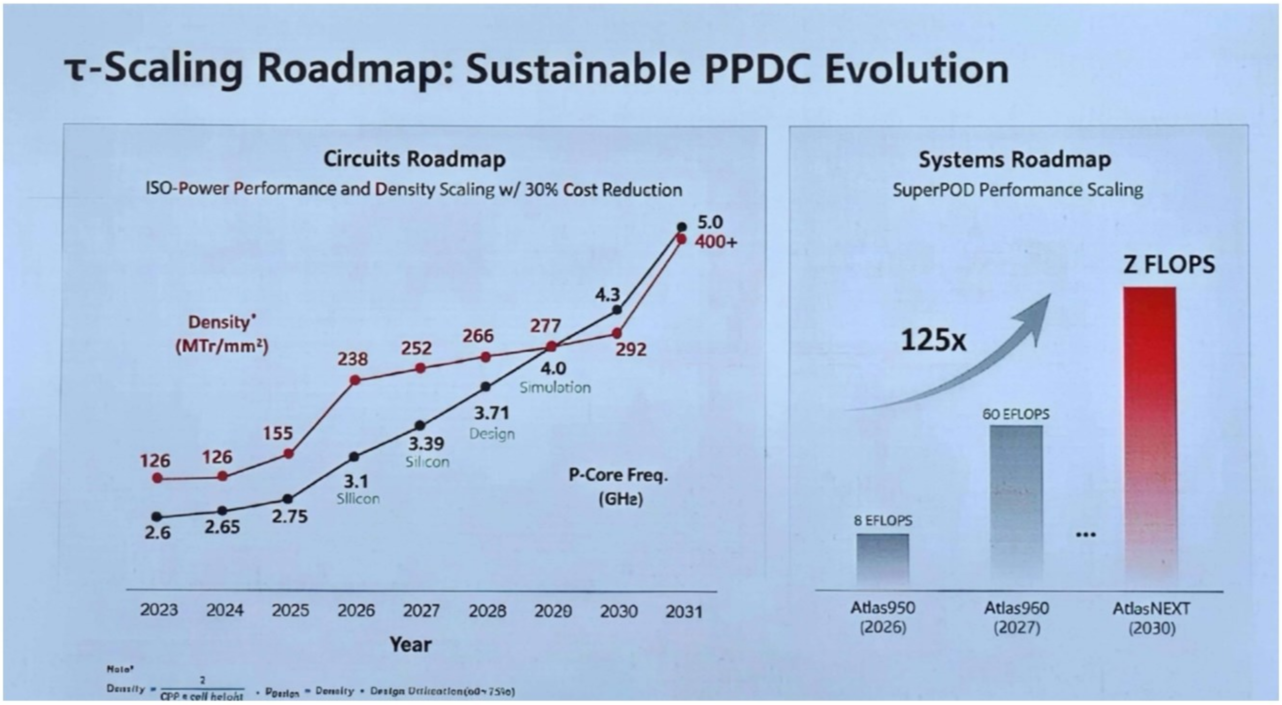
Three key highlights:

1. **Fast transistor density improvement.** Trough advanced packaging (or so-called 'LogicFolding' by Huawei), they are able to improve the transistor density from 155 MTr/mm^2 in 2025 (HiSilicon Kirin 9030, based on SMIC N+3 node, between TSMC N5 and N7) to 238 MTr/mm^2 in 2026 (equivalent to TSMC N3 node density. Likely that's based on two chips stack on top of each other, for example one SMIC N+3 node chip with another SMIC N+2 node chip, so no longer apple to apple comparison). By 2031, they target to achieve 400+ MTr/mm^2 (equivalent to TSMC 14A density, but again not apple to apple comparison);

2. **Steadily increasing max frequency.** The speed of a transistor is generally measured by how fast it can switch, or the frequency, further improvement on the max frequency is achieved through the tau optimization that reduced the latency, which is the critical-path delay between adjacent transistors, which in turn is dominated by interconnect RC;

3. **Exponentially growing AI cluster scaling.** On top of improvement on chip performance, the Tau Scaling Law also emphasizes on the improvement of networking between chips to build exponentially larger AI superPoDs, we have illustrated that in the Huawei AI chip roadmap analysis, and now Huawei targets to achieve 125X improvement in superPoD total compute power by 2030, which indicates an aggressive 3.3X improvement every year.

EXHIBIT 1: **Huawei's tau scaling roadmap**



Source: Company reports, Bernstein analysis

THE LOGICFOLDING INNOVATION

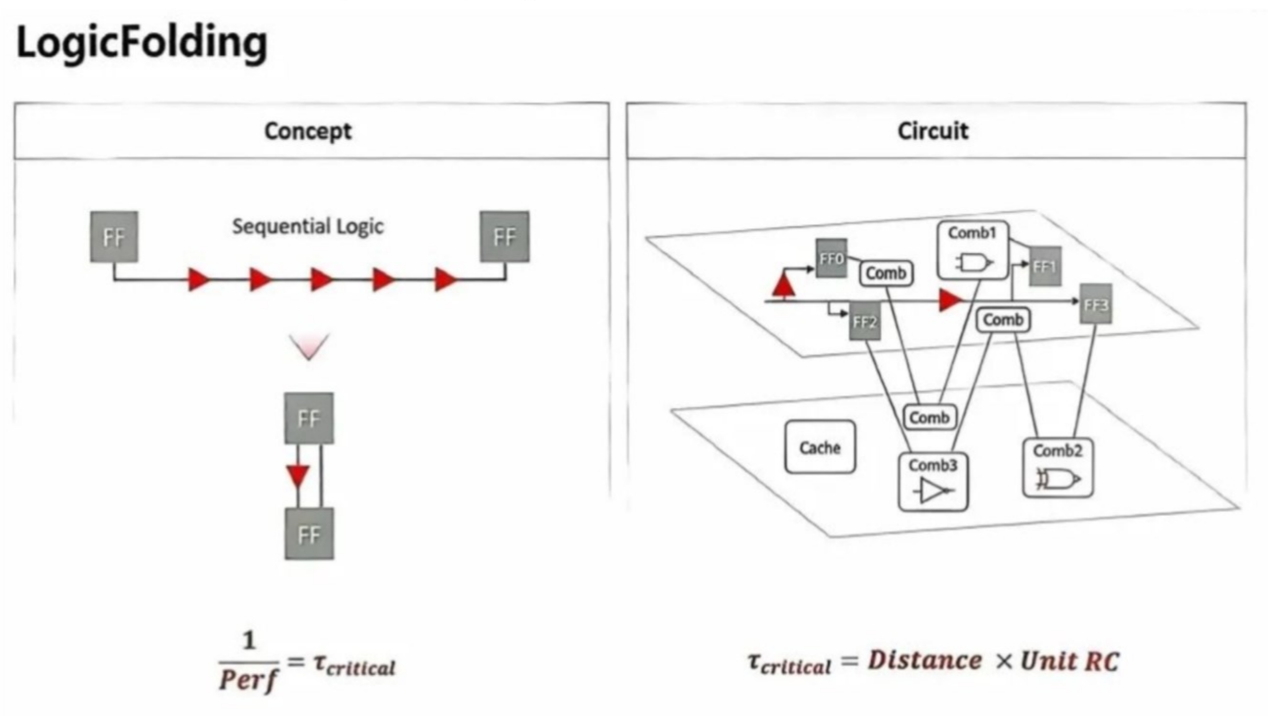
EXHIBIT 2: **Technology improvement in LogicFold**

Sidebar A — *LogicFolding* at a Glance

- Hybrid-bonding pitch: sub-2 µm (1.5 µm in Kirin 2026; target gear ratio ≈ 1)
- Overlay accuracy: under 0.5 µm
- TSV CD/KOZ: sub-1.5 µm; pitch sub-6 µm; failure rate <100 ppm; repair rate 99.9%
- Yield: ~100% with smart redundancy
- Transistor density: 155 →238 MTr/mm² in a single step
- Power-efficiency / frequency gain (SoC P-core): +41% / +13%
- SRAM operating frequency: +40%+
- Clock-buffer count / clock skew / wire length on a representative core: −50% / −25% /−30%

Source: Company reports, Bernstein Analysis

EXHIBIT 3: **Illustration of the LogicFolding principle**



Source: Company reports, Bernstein analysis

THE TIME SCALING THEORY

Link to the full technical paper: A Time Scaling Theory for Multi-Layer Electronic Systems

Copying the abstract here:

For six decades, Moore's geometric scaling drove progress in semiconductors. That industry compact no longer holds: returns from pure dimensional shrinking have flattened, leading-edge design budgets exceed one billion dollars per chip, and cost-per-transistor at the most advanced nodes is no longer falling. This perspective argues for a successor scaling principle — τ scaling — that adopts time itself, rather than transistor area, as the primary metric of progress, applying a single characteristic time constant τ as the unifying optimization target across twelve orders of magnitude, from a switching transistor to a data-center workload. Two production-scale demonstrations are presented. On a mobile SoC, LogicFolding — a methodology that partitions digital, analog, and memory circuits across vertically stacked active tiers — delivers a 55% step-wise increase in transistor density and a 41% power-efficiency gain at a fixed device node. On AI systems, a co-designed stack comprising the memory-semantic Unified Bus fabric, near-packaged Hi-ONE optical I/O, and edge-to-surface 3D Folding projects more than 100× growth in hardware integration by 2035. The deeper claim is methodological: τ scaling is the first scaling principle since Dennard to establish a shared optimization target across the entire computing stack.

The theory dictates optimizing and compressing delay (τ) across four collaborative layers of the compute stack:

- **Transistor Level:** Reducing intrinsic transistor switching and interconnect delays. This part is where the Moore's law has been making the direct impact, where shrinking the feature size of a transistor help to improve the speed of calculation. Beyond shrinking the size of the feature, changing the structure could also improve transistor performance (e.g. from planner to FinFET to GAA structure). We believe local foundry are also making progress at this level, such as **using DUV to manufacture GAA** structure;

- **Circuit Level:** Implementing **LogicFolding** (breaking down and vertically stacking logic, memory, and analog blocks using 3D heterogeneous integration and fine-pitch hybrid bonding) to drastically shorten critical wiring and reduce RC loads. Beyond the traditional 3DIC roadmap, Huawei further bring the design from **'Chip to Chip' stacking to 'Cell to Cell' stacking**, allow different part of the circuit function (the Combinational logic and Sequential logic, which performs the calculation and memory parts within a transistor) to be stacked on to of each other.

- **Chip Level:** Optimizing silicon and software holistically to reduce total execution time. This is commonly cited as '**Design-Technology Co-Optimization**' by the industry. As Huawei is forced to build its own know-how on process, and even build its own fabs, it's also possible that Huawei can move faster in the co-optimization to deliver better results at the chip level.

- **System Level:** Utilizing memory-semantic unified bus structures (e.g., UnifiedBus) and high-speed optical I/O (e.g., Hi-ONE) to bypass physical distance bottlenecks and compress cluster communication latency from tens of microseconds to ~100 nanoseconds. This is the networking architecture optimization illustrated in Huawei's paper on UB-Mesh (UB-Mesh: a Hierarchically Localized nD-FullMesh Datacenter Network Architecture)

Specifically, in the paper Huawei illustrates the advanced packaging technology that they rely on to get to the impressive 238 MTr/mm^2 transistor density in 2026 that likely will be deployed on its next generation mobile phone Huawei Mate90, and the networking innovations that allow Huawei to deliver more scaling capabilities for AI clusters for the next decade.

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